Software Engineering

Assignment 1

Planning Phase

Team Members:

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Campus Sphere

Problem Statement:

THE PROBLEM OF	**challenges in communication, event management, and member coordination within university societies.**
AFFECTS	university society admins, presidents, and members.
THE IMPACT OF WHICH IS	inefficiency in organizing events, managing memberships, and ensuring effective communication.
A SUCCESSFUL SOLUTION WOULD BE	a centralized desktop application, **Campus Sphere**, designed to streamline society management. The system will provide a user-friendly interface for event organization, membership management, and communication, improving efficiency for all stakeholders. Additionally, it will allow for future scalability to accommodate growing needs.

Feature List:

1. **User Management:**
    - Secure login & authentication
    - Role-based access control (Admin, President, Member)
    - User profile management
2. **Society Management:**
    - Create, edit, and delete societies

    - Assign leadership roles (President, VP, etc.)
    - Browse and search for societies
3. **Event Management:**
    - Create, update, and delete events
    - View upcoming events
    - Event notifications for members
4. **Membership Management:**
    - Enroll in a society
    - View society members list
    - Remove inactive members
5. **Feedback & Communication:**
    - Provide feedback on society activities
    - Announcements & notifications system

Product Backlog (User Stories & Acceptance Criteria)

1. User Management

1. **User Story:** As a user, I want to securely log in so that I can access my account.

   Acceptance Criteria:
    - **Given** a user enters valid credentials,
    - **When** they click the login button,
    - **Then** they should be authenticated and redirected to the dashboard.

2. **User Story:** As a user, I want to reset my password so that I can regain access if I forget it.

   Acceptance Criteria:

   - **Given** a user is on the login page,
   - **When** they click "Forgot Password" and enter a registered email,
   - **Then** they should receive a password reset link via email.

3. **User Story:** As an admin, I want to assign roles (Admin, President, Member) so that each user has appropriate access.

   Acceptance Criteria:

   - **Given** an admin is on the user management page,
   - **When** they assign a role to a user and save the changes,
   - **Then** the system should update the user's role accordingly.

4. **User Story:** As a user, I want to edit my profile information so that my details remain up to date.

   Acceptance Criteria:

   - **Given** a user is on the profile settings page,
   - **When** they update their details and save,
   - **Then** the system should store and reflect the new information.

2. Society Management

5. **User Story:** As an admin, I want to create a society so that students can join and participate.

   Acceptance Criteria:

   - **Given** an admin is logged in,
   - **When** they enter society details and submit,

    - **Then** the system should successfully create and store the society.

6. **User Story:** As an admin, I want to edit society details so that information remains updated.

   Acceptance Criteria:

    - **Given** an admin selects a society,
    - **When** they modify its details and save changes,
    - **Then** the updated details should be reflected in the system.

7. **User Story:** As an admin, I want to delete societies so that inactive ones are removed.

   Acceptance Criteria:

    - **Given** an admin selects a society marked for deletion,
    - **When** they confirm the deletion,
    - **Then** the society should be permanently removed from the system.

8. **User Story:** As an admin, I want to assign leadership roles (President, VP) so that societies have proper management.

   Acceptance Criteria:

    - **Given** an admin is on the society settings page,
    - **When** they assign roles to users,
    - **Then** the system should reflect the updated leadership structure.

9. **User Story:** As a user, I want to browse societies so that I can find and join relevant ones.

   Acceptance Criteria:

    - **Given** a user is on the society list page,
    - **When** they search for a society,

- **Then** relevant results should be displayed.

3. Event Management

10. **User Story:** As a president, I want to create events so that my society members are informed about upcoming activities.
    Acceptance Criteria:
    - **Given** a president is logged in,
    - **When** they enter event details and submit,
    - **Then** the system should create the event and notify the members.
11. **User Story:** As a president, I want to edit event details so that I can update information when needed.
    Acceptance Criteria:
    - **Given** a president selects an event,
    - **When** they modify event details and save,
    - **Then** the updated details should be reflected in the system.
12. **User Story:** As a president, I want to delete outdated events so that the event list remains relevant.
    Acceptance Criteria:
    - **Given** a president selects an event,
    - **When** they confirm deletion,
    - **Then** the event should be permanently removed.
13. **User Story:** As a user, I want to view upcoming events so that I can stay informed about society activities.
    Acceptance Criteria:
    - **Given** a user is on the events page,

- **When** they access the event list,
- **Then** all upcoming events should be displayed.

14. **User Story:** As a user, I want to receive event notifications so that I do not miss important activities.

    Acceptance Criteria:

    - **Given** a user is a member of a society,
    - **When** a new event is added,
    - **Then** they should receive a notification.

4. Membership Management

15. **User Story:** As a user, I want to request to join a society so that I can participate in its activities.

    Acceptance Criteria:

    - **Given** a user selects a society,
    - **When** they submit a join request,
    - **Then** the request should be visible to the president for approval.

16. **User Story:** As a president, I want to approve or reject membership requests so that only interested members join.

    Acceptance Criteria:

    - **Given** a president is in the membership requests section,
    - **When** they approve/reject a request,
    - **Then** the member should be added or denied access accordingly.

17. **User Story:** As a president, I want to view my society's member list so that I can manage them efficiently.

    Acceptance Criteria:

- **Given** a president is on the society dashboard,
- **When** they access the members section,
- **Then** they should see a complete and updated list of members.

18. **User Story:** As a member, I want to leave a society so that I am no longer associated with it.

    Acceptance Criteria:

    - **Given** a member is in the society settings,
    - **When** they choose to leave,
    - **Then** their membership should be revoked.

5. Feedback & Communication

19. **User Story:** As a member, I want to provide feedback on my society so that improvements can be made.

    Acceptance Criteria:

    - **Given** a member submits feedback,
    - **When** they enter their comments and submit,
    - **Then** the feedback should be stored and visible to the admin.

20. **User Story:** As an admin, I want to review feedback so that I can analyze society performance.

    Acceptance Criteria:

    - **Given** an admin accesses the feedback section,
    - **When** they filter by society or date,
    - **Then** relevant feedback should be displayed.

21. **User Story:** As a president, I want to send announcements so that my society members stay informed.

    Acceptance Criteria:

    - **Given** a president is logged in,
    - **When** they create an announcement,
    - **Then** it should be sent to all society members as a notification.

22. **User Story:** As a user, I want to view all announcements so that I stay updated.

    Acceptance Criteria:

    - **Given** a user is on the announcements page,
    - **When** they check for updates,
    - **Then** all relevant announcements should be displayed.

23. **User Story:** As an admin, I want to monitor system notifications so that I can ensure communication is effective.

    Acceptance Criteria:

    - **Given** an admin is logged into the system,
    - **When** they access the notification logs,
    - **Then** they should be able to view a list of all sent notifications, including timestamps and recipients.

24. **User Story:** As a user, I want to search for specific announcements so that I can find relevant information.

    Acceptance Criteria:

    - **Given** a user is on the announcements page,
    - **When** they enter a keyword in the search bar,

    - **Then** the system should display announcements containing the keyword.

25. **User Story:** As an admin, I want to track user engagement with notifications so that I can analyze their effectiveness.

    Acceptance Criteria:

    - **Given** an admin is reviewing notification analytics,
    - **When** they access engagement metrics,
    - **Then** the system should display data such as the number of views, clicks, and responses for each notification.

Spring Backlog

1. As a user, I want to securely log in so that I can access my account
2. As a user, I want to browse societies so that I can find and join relevant ones.
3. As a user, I want to view upcoming events so that I can stay informed about society activities.
4. As a user, I want to request to join a society so that I can participate in its activities.
5. As a member, I want to provide feedback on my society so that improvements can be made.

Roles & Responsibilities

- **Product Owner:** Ali Shahzain
- **Scrum Master:** Shabi
- **Scrum Team:** Both members will divide development, testing, and implementation responsibilities equally.

Team Agreement

Methods of Communication:

- **Primary:** WhatsApp (for quick updates and discussions)

- **Secondary:** Email (for formal documentation and progress tracking)
- **Backup:** Phone calls (for urgent matters)

Communication Response Times:

- Messages should be acknowledged within 12 hours.
- Urgent issues (blocking tasks) should be addressed immediately.

Meeting Attendance:

- Weekly virtual meeting (mandatory for both members).
- Additional ad-hoc meetings as needed.
- If a member is unable to attend, they must inform the other at least 24 hours in advance.

Running Meetings:

- Meetings will be held virtually via Google Meet/Zoom.
- Scrum Master (Shabi) will facilitate the meeting.
- Minutes will be documented and shared in a Google Doc.

Meeting Preparation:

- Review previous sprint progress.
- Prepare agenda (issues to discuss, blockers, next sprint goals).

Version Control:

- GitHub Repository for code management.
- Commit Guidelines:
    - Descriptive commit messages (e.g., "Added event scheduling feature").
    - No committing incomplete or broken code.

    - Feature branches must be reviewed before merging into main.

Division of Work:

- Ali Shahzain: Backend development, database management, security.
- Shabi: Frontend development, UI design, testing.
- Tasks will be divided based on sprint goals and availability.
- Stakeholders: University societies, students, and faculty.

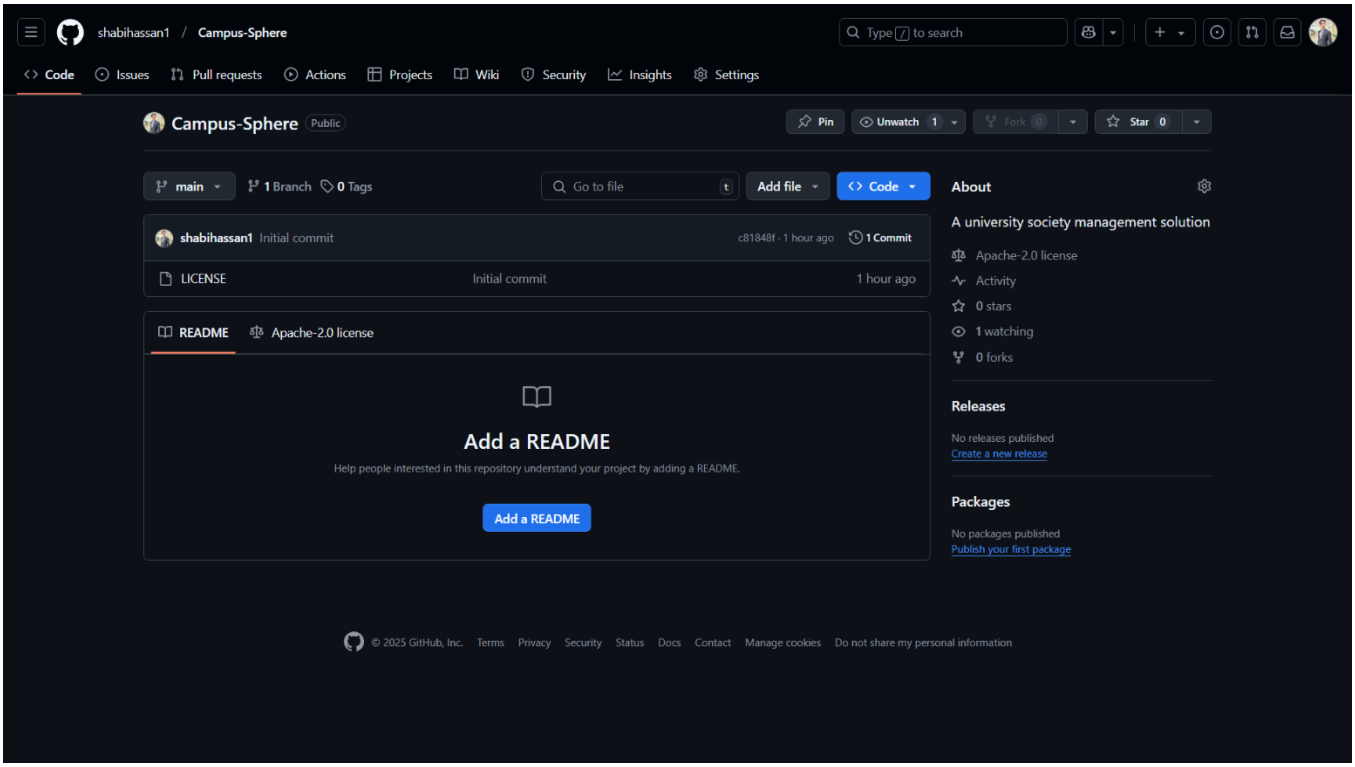
Submitting Assignments:

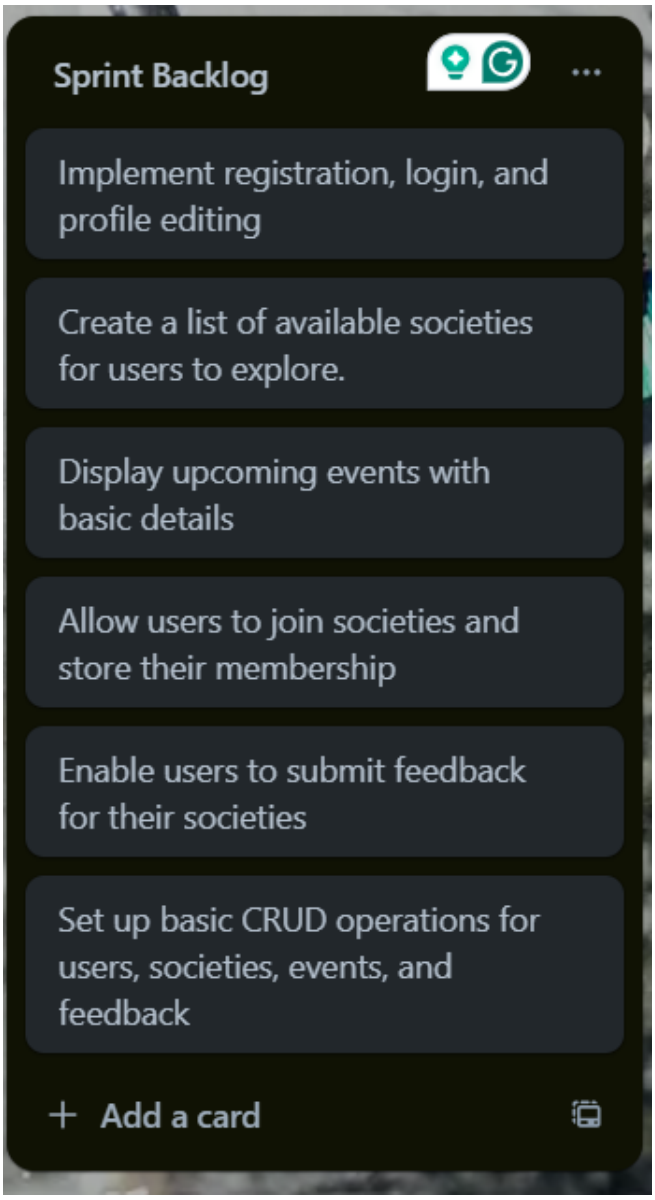
- The final document/code will be reviewed before submission.
- Shabi will submit on Google Classroom after team review.
- Ali Shahzain will verify content completeness before submission.

Contingency Planning:

- If a team member drops out, the remaining member will:
    - Take over unfinished tasks.
    - Request an extension if needed.
- If a team member misses two consecutive meetings, a discussion will be held to resolve the issue.
- Any academic dishonesty will be reported immediately and the responsible person will be removed from task ownership.

GitHub Repository:



https://github.com/shabihassan1/Campus-Sphere

Trello Board Snapshots: